
Bringing R3GAN to Molecule Generation Domain

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Abstract

Generative-Adversarial Networks are known for their problems with mode collapse, which is why they were slowly overtaken with other generative approaches, such as Diffusion Models. R3GAN architecture, proposed in (1), sought to change that, with mathematically mode collapse-robust architecture. In the following report we investigated the applicability of this R3GAN architecture to molecule generation domain, starting from small molecule generation with MolGAN (2), and proceeding to full-scale generation with LatentGAN (3) on MOSES dataset (4). Our findings have shown promising results, as our R3-models were able to beat previous GAN architectures along with EDM [5] Diffusion baseline on all aforementioned tasks in terms of quality score (5).

1 Introduction

The design of new molecules with specific properties is an important part of drug discovery process and materials sciences. Deep generative models that learn from existing data to create novel compounds, avoiding costly manual searches, present promising approach in de novo drug design.

1.1 Related Works

Current deep generative approaches include Generative-Adversarial Networks (GANs), Diffusion Models and Variational Autoencoders. All above-mentioned models can and are utilized in chemical domain problems.

MolGAN (2) was the first state-of-the-art model in small molecule generation. It utilized molecular graphs representation and a GAN architecture. However, it still produced disconnected molecules in some cases and experienced its validity drops when generating larger structures. This later was improved with such models as the Equivariant Diffusion Model (EDM). EDM a diffusion-based framework for 3D molecular generation with a focus on small organic molecules. In this work, we utilized EDM as a baseline.

1.2 R3GAN

Contemporary GAN architectures, such as Wasserstein GAN (WGAN) are trained through a min-max game between generator and discriminator with gradient penalties, expressed as follows:

$$\min_G \max_{D \in \mathcal{D}} \mathbb{E}_{y \sim P_{data}} [D(y)] - \mathbb{E}_{z \sim P_z} [D(G(z))] - \underbrace{\lambda \mathbb{E}_{\hat{y} \sim P_{\hat{y}}} \left[(\|\nabla_{\hat{y}} D(\hat{y})\|_2 - 1)^2 \right]}_{\text{gradient penalty}}.$$

This approach is known for the problem of so-called mode collapse, whereby during training the generator may reach a state where it will always produce a limited set of outputs. In this case, the space in which the generated samples are distributed will be significantly smaller than the original space. A variety of heuristic tricks are typically employed to address this issue.

The authors of R3GAN propose a shift away from this established approach and present an alternative training strategy with three parts:

- Relativistic Paired Loss (R_p): This component generalizes standard GANs by comparing real (y) and generated ($G(z)$) samples jointly: $R_p = \mathbb{E}_{y,z} [D(y) - D(G(z))]$
- Zero-Centered Gradient Penalties R_1 and R_2

$$R_1 + R_2 = \frac{\gamma}{2} (\mathbb{E}_{y,z} [\|\nabla_y D(y)\|_2^2 + \|\nabla_{G(z)} D(G(z))\|_2^2])$$

Which could be expressed as follows:

$$\min_G \max_{D \in \mathcal{D}} \mathbb{E}_{y \sim P_{data} z \sim P_z} \underbrace{[D(y) - D(G(z))]}_{R_p \text{ GAN}} - \underbrace{\frac{\gamma}{2} \mathbb{E}_{y \sim P_{data} z \sim P_z} [\|\nabla_y D(y)\|_2^2 + \|\nabla_{G(z)} D(G(z))\|_2^2]}_{R_1 \text{ and } R_2 \text{ penalties}}.$$

The article formally shows that R3GAN does not have local minima corresponding to mode collapse. Thus, the usage of this approach should help to improve both the training process and samples qualities. In this manuscript, we investigate the applicability of the above approach to the molecule generation domain. Starting on small molecules generation with MolGAN and proceeding with LatentGAN (3).

2 Datasets and Metrics

The first set of experiments was carried on the QM9 dataset. This well-known dataset is curated from the GDB-17 database (166.4B organic molecules) and filtered to 134k stable, synthetically accessible molecules with ≤ 9 heavy atoms (C, O, N, F). Later, to investigate the scalability of the R3-architecture, we used two main frameworks:

- MOSES (4) - filtered subset of ZINC Clean Leads, resulting in 1.93M drug-like molecules. This dataset was used as a training one for LatentGAN.
- NblaDFT - dataset with different conformations for 1.6M drug-like molecules. We used it to train EDM.

3 Evaluation and metrics

In order to evaluate the quality of generation by the models employed, three statistical metrics are applied: validity, which is defined as the fraction of valid molecules in the generated set; novelty - the number of generated molecules that are not presented in the dataset; and uniqueness - the number of unique molecules in the generated set.

For all models we used three standard chemical metrics: QED(6) - quantitative estimation of drug-likeness, synthetic accessibility(7) (SA) score, and LogP. In addition to these, we use a quality metric: the fraction of valid and unique molecules with $\text{QED} \geq 0.6$ and $\text{SA score} \leq 4$, as it was proposed in (5).

Also, for LatentGAN-like model we used metrics from MOSES (4), such W1-Distance between distribution of chemical properties of generated samples and original ones, Fréchet ChemNet Distance, Scaffold/Structure similarity and others.

4 Training

Generally, all models, including final R3MolGAN, R3LatentMolGAN and EDM, were implemented in Pytorch. But to verify the integrity of MolGAN’s Pytorch implementation, the original Tensorflow was used and upgraded with R3.

As R3 presents an pipeline which is pretty dependent on boundary conditions (γ term in regularization), hyperparameter search with optuna was initiated (Fig.1) on 5k subsample of QM9 dataset. The search space for γ was $\gamma \in [0.1, 100]$, for learning rate two separate searches were carried out.

The baseline - EDM model - was trained on a substantial subsample of nablaDFT for comparison with LatentGAN and on a 5k subsample of QM9 for comparison with MolGAN. The hyperparameters were selected to be equivalent to those employed in the original study. During the testing phase, the optimal number of atoms for sampling was empirically determined to be 20.



Figure 1: Optuna search for best hyperparameters for R3MolGAN, logged in wandb. For separate cases, such as $lr \in [5e^{-5}, 2e^{-4}]$ (original R3 article's lr-space) and $lr \in [1e^{-5}, 1e^{-3}]$ different wandb environments were used.

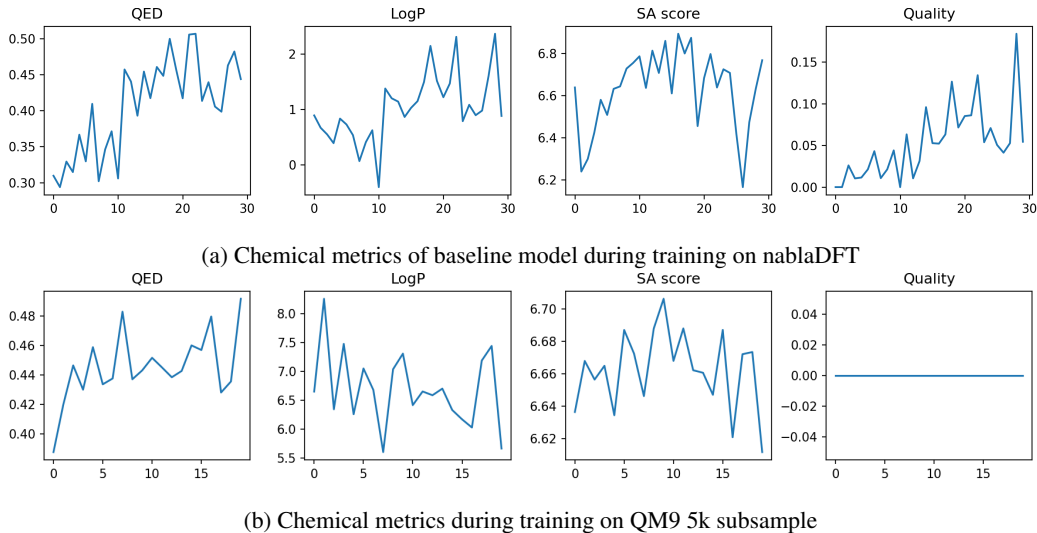


Figure 2: Comparison of chemical metrics during EDM training

5 Results

After Optuna hyperparameter search and final models training, MolGAN-like models and LatentGAN-like models were evaluated and compared against their diffusion counterparts in terms of statistical and chemical metrics. In nearly all of the metrics, R3 models advanced over ordinary GANs. The key values of the optimization problem in terms of generative drug design - quality score values was exceeded by R3GANs in all of the experiments. This, along with lack of necessity for any heuristics in training makes the R3GAN a suitable choice in the domain of molecule generation.

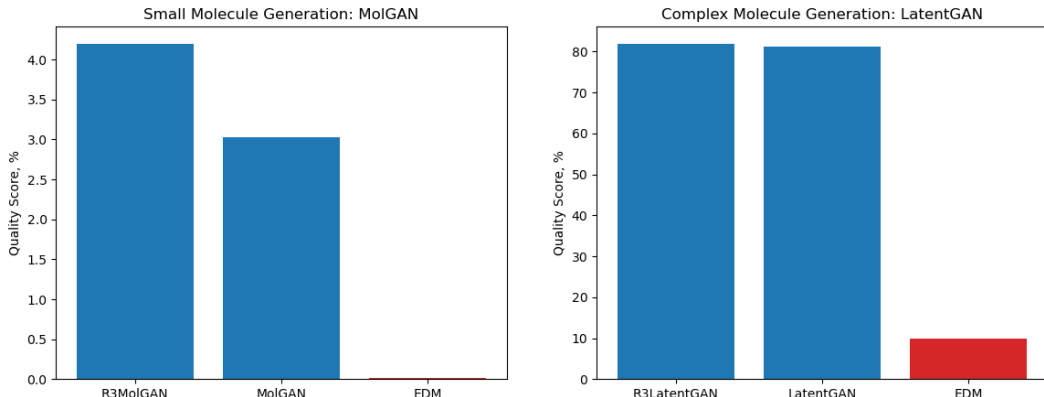


Figure 3: Evaluation of quality scores of generated samples. On the left: quality scores assessment on samples 10000 generated by MolGAN, R3MolGAN and EDM. On the right: quality scores on 15000 samples of LatentGAN, R3LatentGAN and EDM.

Table 1: Comparison of MolGAN, R3MolGAN, and EDM on property-related metrics.

Model	QED Score	SA Score	logP Score	Quality (%)
MolGAN	0.53	0.47	0.33	3.03
R3MolGAN	0.51	0.39	0.38	4.20
EDM	0.49	0.01	0.86	0.01

Table 2: Comparison of LatentGAN and R3-LatentGAN on validity and property-related metrics.

Model	Novelty (%)	Uniqueness (%)	Validity (%)	QED Score	SA Score	logP Score
LatentGAN	95.9	99.79	87.8	0.695	0.613	0.496
R3-LatentGAN	95.9	99.81	87.4	0.703	0.617	0.494

Table 3: MOSES metrics for LatentGAN and R3-LatentGAN. Best values are bolded.

Metric	LatentGAN	R3-LatentGAN
Valid	0.8783	0.8745
Unique@1	1.0000	0.9990
Unique@10	0.9979	0.9981
FCD/Test	0.5387	0.4298
SNN/Test	0.5250	0.5247
Frag/Test	0.9973	0.9985
Scaf/Test	0.8553	0.8409
FCD/TestSF	1.1701	0.9559
SNN/TestSF	0.5012	0.5032
Frag/TestSF	0.9958	0.9972
Scaf/TestSF	0.0655	0.0968
IntDiv	0.8560	0.8578
IntDiv2	0.8497	0.8516
Filters	0.9670	0.9668
W_1 logP	0.0632	0.0780
W_1 SA	0.1123	0.1084
W_1 QED	0.0064	0.0115
Weight	6.1458	4.8171
Novelty	0.9591	0.9589

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